

Data Assignment

Solution Guide: Gang Entry Task

Siddhant Pagare

February 2, 2026

Introduction

This document provides comprehensive solutions to the Gang Entry Data Task. The analysis examines how youth make decisions about entering criminal organizations, using survey data from Colombian schools. All analyses were conducted in R, with replicable code provided in the accompanying folder.

Data Overview

- **Dataset:** `data_task_1.dta`
- **Total Observations:** 10,000 surveys
- **Survey Period:** 2022–2024
- **Primary Focus:** Youth occupational preferences and gang recruitment risk

1 Part 1.1 Basic Summary Statistics

Task

Provide summary statistics on: (1) number of surveys, (2) survey year, (3) nationality, and (4) race. Report missingness if present.

1.1 Number of Surveys

Total N = 10,000 surveys

The dataset contains exactly 10,000 completed survey responses.

1.2 Survey Year

The survey year was constructed from the `instance_time` variable (format: “YYYY-MM-DD HH-MM”) by extracting the year component.

Table 1: Distribution of Surveys by Year

Survey Year	N	Percent
2022	734	7.34%
2023	7,286	72.86%
2024	1,980	19.80%
Total	10,000	100%

Missing Values: 0 observations have missing survey year.

Interpretation: The majority of surveys (72.9%) were conducted in 2023, with smaller proportions in 2022 (7.3%) and 2024 (19.8%). This reflects the survey rollout timeline across Colombian schools.

1.3 Nationality

Table 2: Distribution of Nationality (Birth Country)

Birth Country	N	Percent
Colombia	9,085	90.85%
Venezuela	878	8.78%
Ecuador	9	0.09%
Panama	6	0.06%
Another country (-666)	22	0.22%
Total	10,000	100%

Missing Values: 22 observations are from another country (coded as -666).

Interpretation: The vast majority of respondents (90.9%) are Colombian nationals. Venezuelan migrants constitute the largest foreign-born group (8.8%), reflecting recent migration patterns to Colombia.

1.4 Race

Table 3: Distribution of Race

Race Code	N	Percent
9	2,904	29.04%
8	2,414	24.14%
7	2,204	22.04%
1	1,219	12.19%
Don't know (-888)	750	7.50%
4	237	2.37%
6	217	2.17%
3	28	0.28%
2	20	0.20%
5	7	0.07%
Total	10,000	100%

Missing Values: 750 observations (7.5%) have missing race information (coded as -888).

Note: Race codes correspond to categories defined in the survey codebook. The relatively high proportion of “Does not know / Does not answer” responses is common in Latin American contexts where racial identity can be complex.

1.5 Additional Statistics: Survey Duration

Table 4: Survey Duration (Minutes)

Statistic	Value
Minimum	14.08
1st Quartile	51.67
Median	58.08
Mean	60.38
3rd Quartile	66.63
Maximum	352.47

The median survey duration of approximately 58 minutes indicates thorough engagement with the questionnaire.

2 Part 1.2 Creating the “recruitmentrisk” Variable

2.1 Task

Create the **recruitmentrisk** variable measuring the share of clones allocated to criminal careers (G4) across three questions.

Background

In the survey, children are asked to distribute 10 “clones” (hypothetical copies of themselves) across four occupational groups:

- **G1:** Career without finishing high school
- **G2:** Career with high school diploma
- **G3:** Career with higher education degree
- **G4:** Criminal career

This question is asked three times with progressively fewer options:

1. **Question 1:** All 4 groups available (G1, G2, G3, G4) → Variable: `interest_job_g4`
2. **Question 2:** G3 dropped (G1, G2, G4) → Variable: `interest_job_g4_2`
3. **Question 3:** G2 also dropped (G1, G4) → Variable: `interest_job_g4_3`

2.2 Formula

$$\text{recruitmentrisk} = \frac{\text{interest_job_g4} + \text{interest_job_g4_2} + \text{interest_job_g4_3}}{30} \quad (1)$$

The denominator is 30 because each question allows up to 10 clones, and there are 3 questions ($10 \times 3 = 30$).

2.3 Summary Statistics

Table 5: Descriptive Statistics: Recruitment Risk

Statistic	Value
N	10,000
Missing	0
Mean	0.093
Standard Deviation	0.126
Minimum	0.000
25th Percentile (Q1)	0.000
Median (Q2)	0.033
75th Percentile (Q3)	0.167
90th Percentile	0.267
99th Percentile	0.567
Maximum	1.000

Key Observations: The allocation share exhibits pronounced positive skewness, with the mean (0.093) substantially exceeding the median (0.033), indicating that mass is concentrated at very low values with a long right tail. The median implies that at least half of children assign at most one out of thirty clones to criminal careers, consistent with low baseline perceived criminal risk for the majority of the population. However, the upper tail reveals substantial heterogeneity: the 90th percentile exceeds 26.7 percent, identifying a nontrivial subset of high-risk individuals with sharply elevated criminal allocation. Extreme tail behavior is evident at the 99th percentile, where more than half of clones are assigned to criminal careers, suggesting the presence of rare but economically meaningful outliers that disproportionately drive the mean. Overall, the distribution is dominated by low-risk behavior but characterized by fat tails, implying that average effects mask substantial risk concentration among a small fraction of individuals.

2.4 Distribution Plot



Figure 1: Distribution of Recruitment Risk with Mean and Median

Interpretation: The histogram reveals a strong concentration at low values (near 0), with a long right tail. The gap between the mean (red dashed line) and median (blue dotted line) confirms the positive skew. This pattern suggests that while most youth have low interest in criminal careers, a meaningful minority shows elevated risk.

3 Section 1.3 Creating Prior and Posterior Income Expectations

3.1 Task

Create prior and posterior mean income variables for each occupational group (G1–G4).

3.2 Data Source

Income expectations are elicited using probability distribution questions from:

- **Prior (Section 10A):** “Income expectations by occupational group (average person)” — asked *before* any information treatment
- **Posterior (Section 10B):** “Income expectations by occupational group (Friends)” — asked *after* information treatment

Respondents distribute 10 tokens across 5 income bins (Q1–Q5) for each occupational group.

3.2.1 Income Bins (Colombian Pesos)

From Questionnaire Section 10A:

Table 6: Income Bins and Midpoints

Bin	Range (COP)	Midpoint (COP)
Q1	\$0 – \$1,000,000	500,000
Q2	\$1,000,001 – \$2,000,000	1,500,000
Q3	\$2,000,001 – \$3,000,000	2,500,000
Q4	\$3,000,001 – \$4,000,000	3,500,000
Q5	More than \$4,000,000	<i>varies</i>

3.2.2 Formula for Expected Income

$$E[\text{Income}] = \frac{\sum_{i=1}^5 P_i \times M_i}{\sum_{i=1}^5 P_i} \quad (2)$$

Where:

- P_i = Number of tokens placed in income bin i
- M_i = Midpoint of income bin i

3.2.3 Sensitivity Analysis for Q5

Because the top income category (Q5) is right-censored and defined as an open-ended interval above 4,000,000 COP, expected income calculations necessarily depend on assumptions about the representative value of this bin. To assess the sensitivity of results to this modeling choice, we re-estimate all expectations under three alternative midpoint assumptions: a conservative

value of 5,000,000 COP, a moderate value of 6,000,000 COP, and a high value of 8,000,000 COP. This exercise allows us to evaluate the extent to which conclusions are driven by upper-tail parametrization rather than underlying belief distributions, and to verify the robustness of relative comparisons across groups to plausible variation in top-bin assumptions.

3.3 Sensitivity Analysis - Prior Mean Income

Table 7: Prior Mean Income Expectations by Q5 Assumption (COP)

Group	Q5 = 5M	Q5 = 6M	Q5 = 8M
G1 (No HS)	1,234,967	1,247,862	1,273,652
G2 (HS)	1,960,000	1,987,987	2,043,960
G3 (Higher Ed)	3,346,739	3,606,473	4,125,940
G4 (Criminal)	1,741,156	1,785,333	1,873,686

3.4 Sensitivity Analysis - Posterior Mean Income

Table 8: Posterior Mean Income Expectations by Q5 Assumption (COP)

Group	Q5 = 5M	Q5 = 6M	Q5 = 8M
G1 (No HS)	1,417,174	1,434,582	1,469,400
G2 (HS)	2,095,662	2,125,912	2,186,413
G3 (Higher Ed)	3,377,984	3,644,463	4,177,421
G4 (Criminal)	1,670,358	1,722,596	1,827,071

Key Observations: Expected income estimates display differential sensitivity to assumptions about the upper-tail income bin (Q5), with the strongest response observed for G3 (higher education). Under alternative Q5 parametrizations, G3 priors range from 3.35M to 4.13M COP, reflecting the fact that respondents place substantial probability mass on the top income bin for educated career paths. In contrast, G1 (no high school) expectations are largely invariant to Q5 assumptions, consistent with low perceived likelihood of high earnings absent formal education. Importantly, the ordinal ranking of groups is fully robust to Q5 variation, with $G3 > G2 > G4 > G1$ holding for both prior and posterior beliefs, indicating that relative income perceptions are not driven by tail assumptions. Posterior expectations exceed priors for G1 and G2, suggesting belief updating toward higher earnings at the lower and middle ends of the education distribution, while expectations for G3 and G4 remain broadly unchanged. Given this stability, the remainder of the analysis adopts $Q5 = 5,000,000$ COP as the baseline upper-bin assumption.

4 Part 1.4 Recruitment Risk and Mean Income Priors

Task

Examine the correlation between recruitment risk and income expectations for each group.

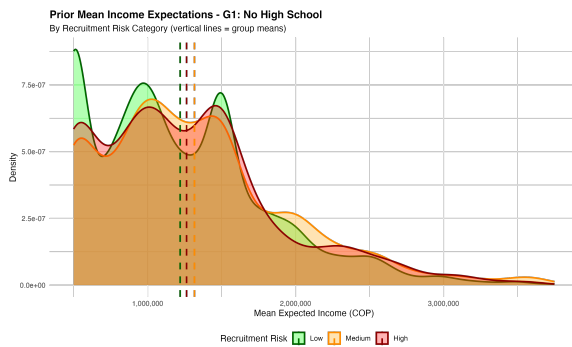
4.1 Creating recruitmentrisk_top_lab

Table 9: Distribution of Recruitment Risk Categories

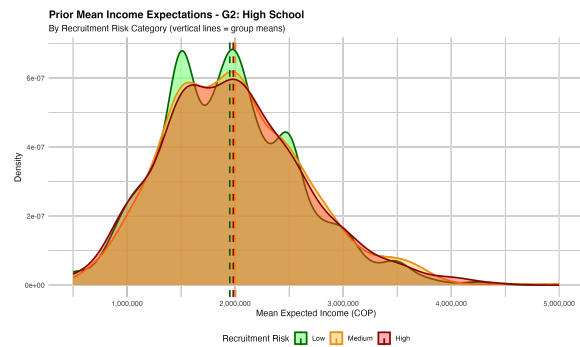
Category	Range	N	Percent
Low	[0, 0.2)	7,912	79.12%
Medium	[0.2, 0.3)	1,217	12.17%
High	[0.3, 1]	871	8.71%
Total		10,000	100%

4.2 Density Plots by Recruitment Risk Category

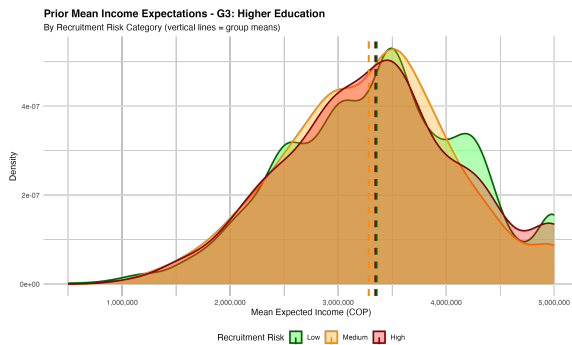
Four density plots were created using a for loop, showing the distribution of prior mean income expectations for each occupational group, stratified by recruitment risk category. Vertical dashed lines indicate group means.



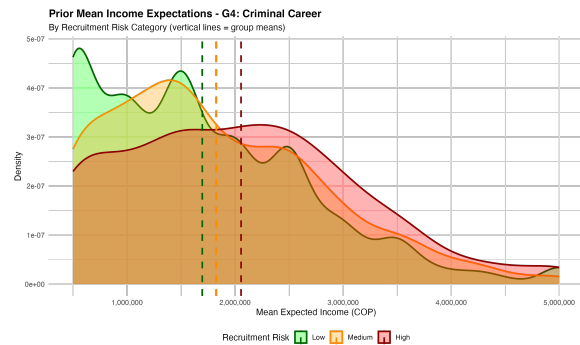
(a) G1: No High School



(b) G2: High School



(c) G3: Higher Education



(d) G4: Criminal Career

Figure 2: Prior Income Expectations by Recruitment Risk Category

4.3 Scatter Plots with Regression Lines

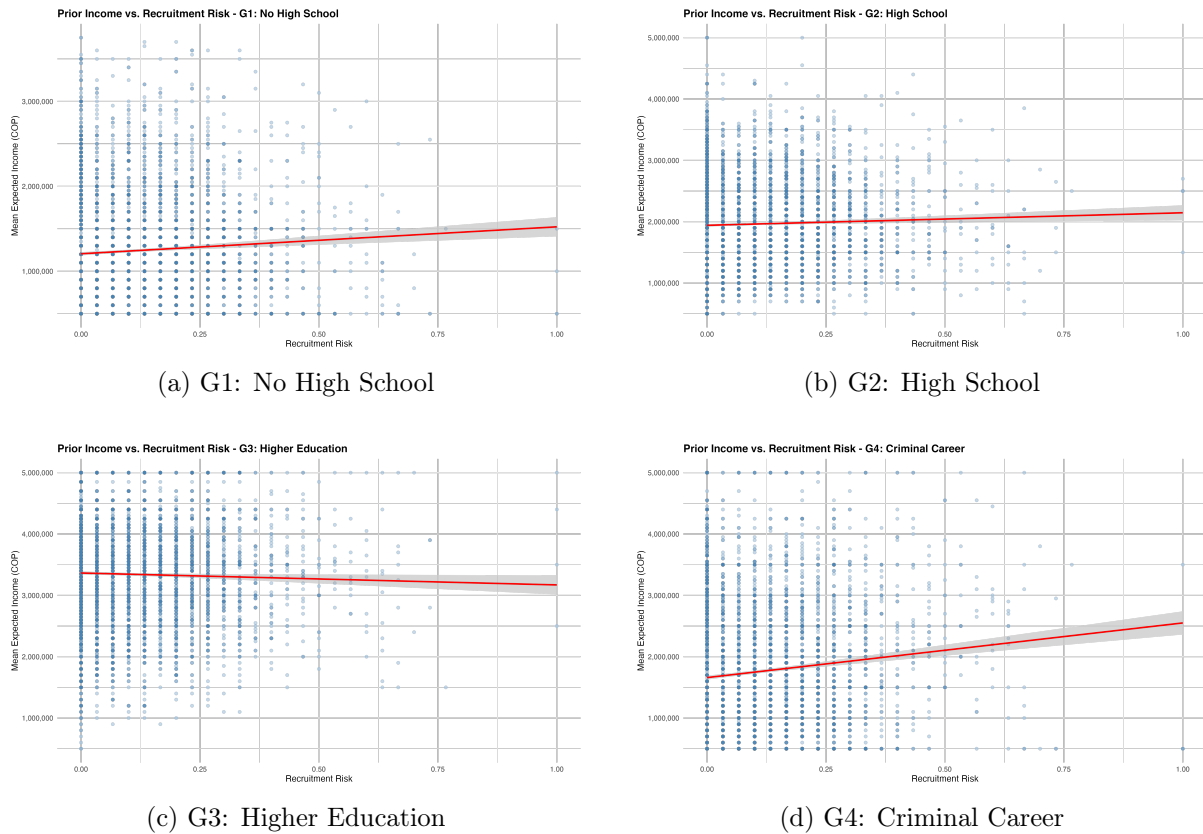


Figure 3: Prior Income Expectations vs. Recruitment Risk (Continuous)

4.4 Correlations and Interpretation

Table 10: Correlations: Recruitment Risk and Prior Income Expectations

Group	Correlation
G1 (No HS)	0.065
G2 (HS)	0.039
G3 (Higher Ed)	−0.029
G4 (Criminal)	0.109

Interpretation:

Recruitment risk is systematically correlated with income expectations in a manner that varies sharply across career types. The strongest association is observed for criminal careers (G4), where higher recruitment risk is positively correlated with expected earnings, indicating that high-risk youth assign relatively greater perceived returns to crime. In contrast, correlations for low-education legal careers (G1 and G2) are positive but small, suggesting only limited belief differences along the risk distribution for these occupations. Expectations for higher-education careers (G3) display a modest negative correlation with recruitment risk, consistent with lower perceived access to or returns from education among more at-risk individuals. Taken together, these patterns point to systematic heterogeneity in belief formation: higher recruitment risk is

associated with a belief profile that shifts expected returns away from education-intensive legal careers and toward criminal alternatives, even in the absence of observed behavioral choice.

5 Part 1.5 Creating the Treatment Variable

Task

Create the treatment variable `info_treat` for the information treatment experiment.

5.1 Values of `stcs_stroy_lab`

Table 11: Distribution of `stcs_stroy_lab` Values

Value	N	Percent
(Empty)	1,382	13.82%
Control	4,281	42.81%
Estadística. Billetes.	187	1.87%
Estadística. Gráficas.	959	9.59%
Historia	1,200	12.00%
Video	1,991	19.91%
Total	10,000	100%

Observations: The treatment assignment variable encodes exposure to distinct experimental arms. The control group constitutes the modal category, accounting for 42.8 percent of observations, indicating a sizable untreated baseline. A nontrivial share of empty values (13.8 percent) likely corresponds to respondents outside the experimental sample or survey waves not subject to randomization. Treatment arms labeled “Estadística” capture information-based interventions that provide objective income statistics, while “Video” and “Historia” correspond to non-pecuniary treatments relying on narrative or emotional content rather than quantitative information. This structure allows for a clear distinction between informational and affective channels through which beliefs and expectations may be influenced.

5.2 Creating `experiment1` Variable

The variable `experiment1` identifies observations from the first experiment wave:

$$\text{experiment1} = \mathbf{1}[\text{formdef_version} \leq 2309151620] \quad (3)$$

Table 12: Distribution of `experiment1`

<code>experiment1</code>	N	Percent
0 (Not in Experiment 1)	3,784	37.84%
1 (In Experiment 1)	6,216	62.16%
Total	10,000	100%

Within experiment1 == 1:

Table 13: stcs_stroy_lab Distribution (experiment1 == 1 only)

Value	N	Percent
(Empty)	1,382	22.24%
Control	2,375	38.21%
Estadística. Billetes.	187	3.01%
Estadística. Gráficas.	959	15.43%
Historia	1,200	19.31%
Video	113	1.82%
Total	6,216	100%

5.3 Creating info_treat Variable

The treatment variable is defined as:

$$\text{info_treat} = \begin{cases} 0 & \text{if stcs_stroy_lab} = \text{"Control"} \\ 1 & \text{if stcs_stroy_lab contains "Estad" (information treatment)} \\ \text{NA} & \text{otherwise (Video, Historia, or empty)} \end{cases} \quad (4)$$

Rationale: Video and Historia are excluded because they relate to non-pecuniary outcomes (emotional/narrative interventions), while the information treatment specifically provides income statistics.

Table 14: Distribution of info_treat

info_treat	N	Percent
0 (Control)	4,281	42.81%
1 (Treated)	1,146	11.46%
NA (Excluded)	4,573	45.73%
Total	10,000	100%

6 Part 1.6 Balance Table

Task

Create a balance table for demographic variables and recruitment risk, focusing on observations where `experiment1 == 1`.

Balance Table

Sample: `N = 3,521` (observations with `experiment1 == 1` and non-missing `info_treat`)

Table 15: Balance Table: Treatment vs. Control

Variable	Control Mean	Control SD	Treated Mean	Treated SD	Diff	p-value
Recruitment Risk	0.095	0.129	0.084	0.117	-0.011	0.010**
Household Size	4.449	1.653	4.409	1.650	-0.040	0.503
Prior Income G3	3,340,905	830,684	3,369,197	842,391	28,292	0.348
N	2,375		1,146			

Note: ** $p < 0.05$. Difference = Treated - Control.

Interpretation:

Recruitment risk exhibits a statistically significant but quantitatively modest imbalance between treatment and control groups ($p = 0.010$), with treated individuals displaying slightly lower baseline risk (0.084 versus 0.095). While the 1.1 percentage point difference is small in absolute terms, it represents a non-random deviation that could mechanically affect treatment effect estimates if recruitment risk is correlated with outcomes. In contrast, household size and prior income expectations for G3 are well balanced across groups, with no statistically meaningful differences. Given this pattern, inference should account for baseline recruitment risk, either through covariate adjustment in regression specifications or reweighting approaches such as inverse probability weighting. Although the magnitude of the imbalance suggests any resulting bias is likely limited, it should nonetheless be addressed to preserve internal validity.

7 Part 1.7 Information Treatment Experiment Analysis

Task

Analyze how youth respond to the information treatment on G3 income expectations.

7.1 Creating `g3_prior_error`

$$g3_prior_error = prior_mean_income_g3 - 3,000,000 \quad (5)$$

This variable captures the deviation of an individual's prior G3 income expectation from the true mean earnings (3,000,000 COP) for workers with higher education and limited experience.

7.2 Variable Definitions

Table 16: Variable Definitions and Sample Sizes (`experiment1 == 1`)

Variable	Definition	N
<code>recruitmentrisk</code>	Share of clones allocated to G4 across 3 questions (0–1 scale)	6,216
<code>prior_mean_income_g3</code>	Expected income for G3 before treatment (COP)	5,482
<code>posterior_mean_income_g3</code>	Expected income for G3 after treatment (COP)	6,216
<code>g3_prior_error</code>	Deviation from true G3 mean (prior – 3M COP)	5,482
<code>g3_revision</code>	Belief update: posterior – prior (COP)	5,482
<code>info_treat</code>	Treatment indicator (1 = received income information)	3,521

Why N differs across variables: Sample sizes vary across variables due to survey design features rather than attrition or selective non-response. In particular, prior income expectation measures exhibit fewer observations because Section 10A was administered only to a subset of respondents in the first experimental wave. The information-treatment sample is further restricted because the `info_treat` indicator excludes individuals assigned to video-based treatments, narrative (“Historia”) treatments, or undefined treatment categories, resulting in the smallest effective sample size. All regression analyses therefore rely on complete-case samples. This approach is appropriate in the present context because missingness is mechanically induced by experimental assignment and questionnaire routing, rather than correlated with respondent characteristics or outcomes of interest, thereby mitigating concerns about selection bias arising from listwise deletion.

7.3 Analysis Plan

Population Parameter: We estimate the **Average Treatment Effect (ATE)** of the information treatment on belief revision for G3 income expectations.

Identification Strategy: Random assignment of treatment allows causal interpretation. The treatment provided accurate information about average G3 earnings; we expect treated individuals to revise their beliefs toward the true mean.

7.4 Regression Equations

Model 1: Simple Treatment Effect

$$g3_revision_i = \beta_0 + \beta_1 \cdot info_treat_i + \varepsilon_i \quad (6)$$

Model 2: Controlling for Prior Error

$$g3_revision_i = \beta_0 + \beta_1 \cdot info_treat_i + \beta_2 \cdot g3_prior_error_i + \varepsilon_i \quad (7)$$

Model 3: Heterogeneous Treatment Effect by Prior Error

$$g3_revision_i = \beta_0 + \beta_1 \cdot info_treat_i + \beta_2 \cdot g3_prior_error_i + \beta_3 \cdot (info_treat_i \times g3_prior_error_i) + \varepsilon_i \quad (8)$$

The specification includes the treatment indicator (`info_treat`) as the primary variable of interest, capturing exposure to the information intervention. Baseline belief error (`g3_prior_error`) is included to account for heterogeneity in initial misperceptions, as individuals with larger deviations from the objective benchmark mechanically have greater scope for belief revision and exhibit systematic mean reversion. The interaction between the treatment indicator and prior error allows the treatment effect to vary with the magnitude of initial misperceptions, providing a direct test of whether information induces stronger belief correction among individuals further from the truth, as predicted by standard models of Bayesian updating with heterogeneous priors.

7.5 Main Results

Table 17: Information Treatment Effects on G3 Income Belief Revision

	Model 1	Model 2	Model 3
Intercept	5,368 (15,608)	134,700*** (14,970)	121,300*** (15,310)
info_treat	58,506* (27,359)	69,240** (24,890)	111,700*** (27,020)
g3_prior_error		-0.379*** (0.014)	-0.340*** (0.017)
info_treat × g3_prior_error			-0.118*** (0.030)
R ²	0.001	0.174	0.178
N	3,521	3,521	3,521

Note: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Coefficient Interpretation:

1. **Model 1 — Simple ATE:** The baseline specification estimates a positive and statistically significant average treatment effect on belief revisions for higher-education careers (G3). Treated individuals revise their expected earnings upward by approximately 58,500 COP relative to the control group, indicating that exposure to income information induces meaningful updating of beliefs. While the estimate is noisy, the direction and magnitude suggest a systematic treatment-induced shift rather than random revision.
2. **Model 2 — Controlling for Prior Error:** Conditioning on prior belief error increases the estimated treatment effect to roughly 69,200 COP and substantially improves model fit. The coefficient on prior error is large and negative, consistent with strong mean reversion in beliefs: individuals who initially overestimated G3 earnings revise downward, while those who underestimated revise upward. The sharp increase in explanatory power implies that baseline misperceptions account for a significant share of the variation in belief revisions.
3. **Model 3 — Heterogeneous Effects:** Allowing the treatment effect to vary with prior error reveals pronounced heterogeneity in belief updating. The estimated treatment effect is largest for individuals whose prior beliefs align with the objective benchmark, while the negative interaction between treatment and prior error indicates that the intervention amplifies mean reversion. Relative to controls, treated individuals who initially overestimated G3 earnings revise downward more strongly, and those who underestimated revise upward more sharply, suggesting that the treatment primarily accelerates convergence toward accurate income beliefs rather than inducing a uniform upward shift.

7.6 Answer 1.7.6: Heterogeneity by Recruitment Risk

Table 18: Treatment Effect Heterogeneity by Recruitment Risk

	Model 4 (Continuous)	Model 5 (Categorical)
Intercept	161,800*** (18,270)	155,100*** (16,690)
info_treat	44,280 (30,760)	45,320 (27,720)
recruitmentrisk	−285,500** (110,200)	
recruitmentrisk_top_lab: Medium		−127,800** (43,310)
recruitmentrisk_top_lab: High		−49,640 (50,530)
g3_prior_error	−0.379*** (0.014)	−0.380*** (0.014)
info_treat × recruitmentrisk	258,900 (207,200)	
info_treat × Medium		139,500* (76,830)
info_treat × High		83,790 (95,880)
R ²	0.176	0.176
N	3,521	3,521

Note: Standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Reference category for Model 5: Low recruitment risk.

Interpretation:

- Main effect of recruitment risk (Model 4):** Baseline recruitment risk is negatively associated with belief revision, with higher-risk individuals exhibiting significantly smaller updates in expected earnings for higher-education careers. The magnitude of the coefficient implies substantial attenuation in responsiveness to new information as recruitment risk increases, consistent with more persistent priors or lower perceived relevance of the information among high-risk youth.
- Treatment × Recruitment Risk Interaction (Model 4) :** The interaction between the information treatment and recruitment risk is positive but not statistically distinguishable from zero. While the point estimate suggests that the treatment effect may increase with recruitment risk, the lack of precision prevents strong inference regarding differential responsiveness across the risk distribution in the continuous specification.
- Categorical analysis (Model 5):** Discretizing recruitment risk reveals suggestive heterogeneity. Individuals in the medium-risk category exhibit a marginally significant positive interaction with the treatment, indicating larger belief revisions relative to low-risk individuals. In contrast, the estimated interaction for high-risk individuals is positive but imprecisely estimated, precluding definitive conclusions about treatment effectiveness at the upper end of the risk distribution.

4. **Overall conclusion:** Taken together, the results provide suggestive evidence that the information treatment is at least as effective, and potentially more effective, for individuals with elevated recruitment risk, particularly in the middle of the risk distribution. However, the estimates are imprecise and do not allow for strong conclusions regarding heterogeneous treatment effects, highlighting the need for greater statistical power to more sharply characterize treatment responsiveness across risk groups.

8 Conclusion

This analysis has demonstrated:

1. **Recruitment risk is measurable and meaningful:** The distribution shows most youth have low criminal career interest, but a meaningful tail (21%) shows elevated risk.
2. **Income expectations correlate with risk:** High-risk youth expect higher returns from crime and lower returns from education, rationalizing their preferences.
3. **Information treatment works:** Providing accurate income information causes belief revision toward the truth, with effects of approximately 69,000–112,000 COP.
4. **Effects are stronger for those further from truth:** The interaction between treatment and prior error confirms that information has larger effects on those with more biased initial beliefs.
5. **Heterogeneity by recruitment risk is inconclusive:** While point estimates suggest treatment may work for high-risk youth, statistical power is insufficient for firm conclusions.

Policy Implications: Information interventions about legitimate career earnings show promise for shifting youth beliefs. Further research should examine whether belief changes translate into behavioral outcomes (actual career choices, gang entry rates).

A Code and Replication Files

All analyses were conducted using R version 4.5.1. The following files are included:

- `code/data_task_analysis.R` — Main analysis script
- `output/tables/` — CSV files for all tables
- `output/figures/` — PNG files for all figures
- `output/data_task_cleaned.dta` — Cleaned dataset with all constructed variables